Higher Quadratic Spectral Jets

Harmonic–Stieltjes factorization defects,
polylogarithmic second derivatives, and exact mass collisions

A research continuation prepared for Vladimir Reshetnikov's ProveIt programme
Mathematical derivations and computational audit with OpenAI assistance

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Abstract

The first derivative of a zeta-regularized quadratic spectrum can admit an anomaly-free factorization even when none of its higher derivatives does. We determine the exact distinction. For every polynomial weight and every spectral derivative order, an explicit harmonic-number coefficient formula separates normally convergent sums from Hurwitz derivatives and generalized Stieltjes constants. A finite local term records the collision with the Hurwitz pole. Even weights annihilate all such local residues, but no nonzero polynomial weight annihilates a fixed higher-order defect for every positive mass. A joint meromorphic formula explains the discontinuous first-derivative behavior at noninteger-weight resonances.

For the bilateral quadratic spectrum we evaluate the second derivative at every nonpositive integer using finitely many polylogarithms, their order derivatives, and one explicitly defined convergent polylogarithm integral family. A Gamma normalization then gives an exact mass-antiderivative ladder at every spectral order, with complete harmonic coefficients and fixed boundary constants. For two distinct quadratic masses we give the second derivative for an arbitrary polynomial weight, its fully explicit polynomial subtraction, an ordinary one-dimensional polylogarithmic integral, and a convergent collision expansion whose every coefficient is a finite polygamma expression. A separate logarithmic-square identity provides Gamma-weighted integrals and a dilogarithmic antiderivative.

These results address four concrete extension questions in the preceding *Quadratic-Spectrum Stieltjes Identities* report. The classical multiplicative-anomaly and Bessel ingredients retain their attribution. No minimality of the added integral coordinate, numerical period independence, or resolution of the canonical Gaussian conjectures is asserted. The delivery includes all-index proofs, 834 finite exact assertions, five rejected mutation controls, 63 independent numerical comparisons, source provenance, and additive integration notes.

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Contents

1 Research target, conventions, and logical scope **3**
1.1 Four questions and their answers . . . 3
1.2 Notation and branches . . . 3

2 All-order factorization defects: local residues and convergent terms **4**
2.1 An explicit finite-subtraction operator . . . 4
2.2 The first two orders, and what parity really proves . . . 5

3 A harmonic–Stieltjes coefficient formula at every derivative order **7**
3.1 Concrete consequences . . . 8

4 Noninteger weights and the joint geometry of resonance **9**

5 Logarithmic squares, Gamma integrals, and a dilogarithmic primitive **10**
5.1 An anchored primitive with explicit dilogarithms . . . 11

6 Every negative-integer second jet of the bilateral quadratic spectrum **12**
6.1 An explicit half-integer order derivative . . . 13
6.2 One explicit polylogarithm integral family . . . 13
6.3 The complete second-jet formula . . . 14

7 Harmonic normalization and all anchored mass antiderivatives **15**
7.1 Explicit second-order primitives . . . 16

8 Two quadratic masses: the exact second derivative and collision calculus **17**
8.1 The complete subtraction, not a formal divergent logarithm sum . . . 17
8.2 An ordinary one-dimensional polylogarithmic reduction . . . 19
8.3 Every collision coefficient is a finite polygamma expression . . . 20
8.4 An exact averaged logarithmic-square identity . . . 21

9 Source audit, mathematical status, and reproducible evidence **21**
9.1 What was actually inspected . . . 21
9.2 Corrections to tempting but invalid extensions . . . 22
9.3 Executed finite exact checks . . . 22
9.4 Executed numerical diagnostics . . . 22
9.5 Replay and additive integration . . . 23

10 Further research questions and concrete next targets **23**

Conclusion **24**

A Independent Hurwitz expansions used in the verification **25**
A.1 The second jet at a negative integer . . . 25
A.2 The two-mass second derivative without using the defect . . . 25

B Selected symbolic and numerical receipts **26**

References **26**

1 Research target, conventions, and logical scope

The preceding report *Quadratic-Spectrum Stieltjes Identities* [1] proves a first-derivative factorization formula for polynomially weighted quadratic spectra. It classifies the weights whose local correction vanishes identically in the mass, and obtains finite polylogarithmic formulas at all nonpositive integer spectral points. Its final research agenda asks for higher spectral derivatives, higher-jet factorization defects, noninteger-weight resonances, and the second derivative of a product with distinct masses. Those questions, rather than another first-derivative rederivation, are the starting point of this article.

The canonical manuscript already contains a large system of Hurwitz–Stieltjes and Gamma identities [2]. Its source inventory and incoming collection are too extensive for a claim that every preserved result was audited here. The contribution below is an explicit, proved continuation of the inspected predecessor. Some ingredients are classical: the Hurwitz Laurent expansion, Gamma and polygamma identities, Poisson summation for a quadratic spectrum, half-integer Bessel formulas, and the first-derivative multiplicative-anomaly mechanism [3, 5, 6, 7, 9, 10]. We do not claim historical priority for those ingredients or for every specialization of the resulting identities.

1.1 Four questions and their answers

The predecessor's question on higher-jet defects is answered by Theorems 2.1 and 3.1, with a complete vanishing classification in Theorem 2.3. Its question on noninteger polynomial degree receives a joint meromorphic answer and an exact finite-part conversion in Theorem 4.1.

The higher-negative-integer question is answered constructively at the *second* derivative by Theorem 6.2. This specifies an adequate explicit function algebra, but does not show it is the smallest possible one. The two-mass second-derivative and collision question is answered for every polynomial weight in Theorems 8.1–8.3. Higher spectral orders have a complete factorization-defect calculus and mass-transport law here, not a claim of a finite Fourier reduction at every order.

1.2 Notation and branches

Throughout the one-sided statements, $a, b > 0$ and $x_n = n + a$, $n \geq 0$. A polynomial $W(x) = \sum_{d=0}^{D} w_d x^d$ may have complex coefficients. For bilateral statements, $0 < a < 1$ and the lattice is $x = n + a$, $n \in \mathbb{Z}$. The positive squared masses are denoted $u = b^2$ and $v = c^2$.

All logarithms of positive numbers are real. On the right half-plane we use the principal analytic logarithm, so for $x > 0$

$$\log(x^2 + b^2) = \log(x + ib) + \log(x - ib).$$

A logarithm of a quotient in a complex-shift identity means the *difference* of the specified logarithms; we do not silently change branches by combining them. Parameter continuation is always from these domains. The exponential arguments of polylogarithms in the massive formulas lie strictly inside the unit disk.

We write $\zeta^{(r)}(s, a) = \partial_s^r \zeta(s, a)$ for order derivatives, not derivatives in a, and use the standard Stieltjes convention

$$\zeta(1 + t, a) = \frac{1}{t} + \sum_{r \geq 0} \frac{(-1)^r \gamma_r(a)}{r!} t^r, \qquad \gamma_0(a) = -\psi(a). \tag{1.1}$$

The generalized harmonic numbers are

$$H_m^{(r)} = \sum_{k=1}^{m} k^{-r}, \qquad H_m = H_m^{(1)}, \qquad H_0^{(r)} = 0.$$

The complete exponential Bell polynomials Y_r are defined by

$$\exp\left(\sum_{j\ge1}\kappa_j\frac{t^j}{j!}\right)=\sum_{r\ge0}Y_r(\kappa_1,\dots,\kappa_r)\frac{t^r}{r!}. \tag{1.2}$$

A superscript [1] on Li_p means $\partial_p\,\mathrm{Li}_p$, whereas $\psi^{(j)}$ denotes the ordinary jth derivative of digamma.

The notation $[x^{-1}]$ extracts the coefficient in the expansion at positive infinity. A finite part is never an unspecified summation method: the regularizing coordinate and the subtraction are displayed. All new numerical comparisons are floating-point diagnostics. The proofs, not those comparisons, establish the all-index statements.

2 All-order factorization defects: local residues and convergent terms

Define initially in their absolute-convergence half-planes

$$Z_W(s;a,b)=\sum_{n\ge0}W(x_n)(x_n^2+b^2)^{-s}, \tag{2.1}$$

$$F_{W,\pm}(s;a,b)=\sum_{n\ge0}W(x_n)(x_n\pm ib)^{-s}. \tag{2.2}$$

Their initial conditions are respectively $\mathrm{Re}\,s>(D+1)/2$ and $\mathrm{Re}\,s>D+1$. Introduce the common-scale difference

$$\mathcal{D}_W(s)=\tfrac12\big(F_{W,+}(2s)+F_{W,-}(2s)\big)-Z_W(s), \tag{2.3}$$

$$\Delta_r(W;a,b)=\mathcal{D}_W^{(r)}(0)=2^{r-1}\big(F_{W,+}^{(r)}(0)+F_{W,-}^{(r)}(0)\big)-Z_W^{(r)}(0). \tag{2.4}$$

The factors of two are part of the definition and are essential from second order onward.

Set

$$\ell_\pm(x)=\log(1\pm ib/x),\qquad L(x)=\log(1+b^2/x^2),\qquad D_b(x)=\ell_+(x)-\ell_-(x)=2i\arctan(b/x). \tag{2.5}$$

The letter D_b is a logarithmic difference, not the degree of W. Pointwise, the summand of (2.3), after removing $W(x)x^{-2s}$, is

$$e^{-sL(x)}\big(\cosh(sD_b(x))-1\big)=\sum_{k\ge2}h_k(x)s^k, \tag{2.6}$$

where

$$h_k(x)=\sum_{q=1}^{\lfloor k/2\rfloor}\frac{D_b(x)^{2q}(-L(x))^{k-2q}}{(2q)!(k-2q)!}. \tag{2.7}$$

In particular,

$$h_2=\tfrac12D_b^2,\qquad h_3=-\tfrac12LD_b^2,\qquad h_4=\tfrac14L^2D_b^2+\tfrac{1}{24}D_b^4.$$

Only even powers of x^{-1} occur in every h_k.

2.1 An explicit finite-subtraction operator

For fixed k, let

$$S_k(x)=\sum_{e\ge-1}a_{k,e}x^e \tag{2.8}$$

be the finite part of the expansion of $W(x)h_k(x)$ at infinity containing all exponents at least -1. Put $R_k = a_{k,-1}$, with missing coefficients understood to be zero. Then $W(x)h_k(x) - S_k(x) = O(x^{-2})$. Define

$$\mathcal{H}_k(t) = \sum_{n\geq 0}(W(x_n)h_k(x_n) - S_k(x_n))x_n^{-t} + \sum_{e\geq -1} a_{k,e}\zeta(t-e,a), \tag{2.9}$$

$$\mathcal{H}_{k,\mathrm{reg}}(t) = \mathcal{H}_k(t) - R_k/t. \tag{2.10}$$

The series is normally convergent for $\operatorname{Re} t > -1$. In a neighborhood of zero the second line is holomorphic. This construction fixes the regularizing coordinate to be $x_n = n + a$.

Theorem 2.1 (Complete local/nonlocal decomposition)**.** *The functions in* (2.1) *and the two linear-factor functions are holomorphic at $s = 0$ after meromorphic continuation. For every $r \geq 1$,*

$$\boxed{\frac{\Delta_r(W;a,b)}{r!} = \frac{R_{r+1}}{2} + \sum_{k=2}^{r} \frac{2^{r-k}}{(r-k)!}\mathcal{H}_{k,\mathrm{reg}}^{(r-k)}(0).} \tag{2.11}$$

Every regular derivative in this formula has the ordinary convergent representation

$$\begin{aligned}\mathcal{H}_{k,\mathrm{reg}}^{(j)}(0) = (-1)^j \sum_{n\geq 0}(W(x_n)h_k(x_n) - S_k(x_n))\log^j x_n \\ + \sum_{e\neq -1} a_{k,e}\zeta^{(j)}(-e,a) + a_{k,-1}(-1)^j\gamma_j(a).\end{aligned} \tag{2.12}$$

The coefficients $a_{k,e}$ are finite polynomials in b and the coefficients of W. No unperformed divergent sum is present on the right.

Proof. Choose a finite head so that the remaining x_n exceed $2b$. On the tail, the inverse-x expansions of the logarithms and exponentials converge uniformly on compact sets of s near zero. Subtract their terms through inverse degree $D+1$. The resulting summand is locally uniformly $O(x_n^{-2+2|\operatorname{Re} s|})$, up to a fixed polynomial in $\log x_n$ after any fixed number of derivatives. It is therefore normally summable for $|\operatorname{Re} s| < 1/4$. The subtracted terms are finite multiples of $\zeta(2s + j - d, a + N)$.

The only possible pole at zero is the term with $j - d = 1$. For each individual spectrum its inverse-power coefficient vanishes at least linearly in s, canceling that simple pole. The degree-zero coefficient only gives Hurwitz values at nonpositive integers and is regular. The finite head is entire in s. This proves holomorphy of the individual spectra at zero.

For the difference, (2.6) starts at s^2. The same subtraction argument allows coefficient extraction from $\sum_{k\geq 2} s^k\mathcal{H}_k(2s)$. Since

$$\mathcal{H}_k(2s) = \frac{R_k}{2s} + \sum_{j\geq 0}\frac{2^j\mathcal{H}_{k,\mathrm{reg}}^{(j)}(0)}{j!}s^j,$$

the coefficient of s^r is exactly (2.11). Only $k \leq r+1$ can contribute. The finite head and the tail can be recombined in (2.9); its ordinary remainder converges even when $b > a$. Finally differentiating that normally convergent remainder and using (1.1) gives (2.12). □

2.2 The first two orders, and what parity really proves

At the first order, Theorem 2.1 recovers the predecessor's local formula:

$$\begin{aligned}\Delta_1(W;a,b) &= \tfrac{1}{4}[x^{-1}]W(x)D_b(x)^2 \\ &= \sum_{j\geq 1} w_{2j-1}\frac{(-1)^j b^{2j}}{j}\left(H_{2j} - \tfrac{1}{2}H_j\right).\end{aligned} \tag{2.13}$$

The sum is finite and independent of a. Its vanishing for every $b > 0$ is equivalent to W being even. For example

$$\Delta_1(1) = 0, \qquad \Delta_1(x) = -b^2, \qquad \Delta_1(x^3) = \tfrac{2}{3}b^4.$$

These are cited predecessor results, also recovered here.

For the second derivative we obtain instead

$$\Delta_2(W) = \mathrm{Reg}_x \sum_{n\geq 0} W(x_n) D_b(x_n)^2 - \tfrac{1}{2}[x^{-1}]W(x)L(x)D_b(x)^2, \tag{2.14}$$

where Reg_x is precisely (2.9) with the constant Laurent term extracted at $t = 0$. Evenness of W removes every R_k, because an even inverse-power expansion cannot contain x^{-1}. It does *not* remove the regular parts in (2.11).

Corollary 2.2 (An ordinary convergent second defect)**.** *For every $a, b > 0$,*

$$\boxed{\Delta_2(1; a, b) = -4 \sum_{n=0}^{\infty} \arctan^2\left(\frac{b}{n+a}\right).} \tag{2.15}$$

In particular this defect is strictly negative.

Proof. For $W = 1$, $D_b^2 = O(x^{-2})$, so no subtraction or residue occurs. Insert $D_b = 2i \arctan(b/x)$ in (2.14). □

The sign is used to give an exact counterexample, not as the principal research target. The predecessor correctly warned against higher-jet factorization; (2.15) makes its missing term explicit.

Theorem 2.3 (Universal higher-order rigidity)**.** *Fix $a > 0$ and an integer $r \geq 2$. For $W \in \mathbb{C}[x]$,*

$$\Delta_r(W; a, b) = 0 \quad \textit{for every } b > 0 \quad \Longleftrightarrow \quad W = 0.$$

Thus even weights are the exact first-order kernel, but the kernel at each fixed higher order is trivial when vanishing is required for all positive masses.

Proof. Suppose $W \neq 0$. There is an $n \geq 0$ with $W(x_n) \neq 0$. Continue the finite-head/tail expression in the complex b-plane along a path from the positive real axis to a punctured neighborhood of $b_0 = ix_n$, avoiding the other points $\pm ix_k$. Choose the head large enough that the remaining tail has $x_k > |b_0| + 1$. Its uniformly convergent inverse-power representation is holomorphic near b_0. All other head terms are also holomorphic there on the continued local branches.

For the nth term, $x_n + ib$ has a simple zero at b_0, while $x_n - ib$ is nonzero. Formula (2.4) gives the coefficient of the highest logarithmic power $\log^r(b - b_0)$ as

$$(-1)^r(2^{r-1} - 1)W(x_n),$$

which is nonzero. Lower powers of that logarithm and holomorphic terms cannot cancel it. If the original function vanished on the entire positive real axis, its analytic continuation would vanish along the chosen path by the identity theorem, contradicting this local singularity. The converse for $W = 0$ is immediate. □

This is an analytic vanishing theorem for a specified family. It is not a theorem about algebraic independence of numerical special values.

3 A harmonic–Stieltjes coefficient formula at every derivative order

The local/nonlocal theorem is global and coordinate-explicit. A second form turns every derivative into a normally convergent Hurwitz series with finite rational harmonic coefficients. It is often more convenient for symbolic work and high-precision evaluation.

For $j \geq 1$ define

$$C_j(s) = \frac{(2s)_{2j}}{(2j)!} - \frac{(s)_j}{j!} = \sum_{k=2}^{2j} c_{j,k} s^k. \tag{3.1}$$

The two linear terms cancel. In product form,

$$C_j(s) = \frac{s}{j} \left\{ \prod_{h=1}^{2j-1} \left(1 + \frac{2s}{h}\right) - \prod_{h=1}^{j-1} \left(1 + \frac{s}{h}\right) \right\}. \tag{3.2}$$

Consequently every $c_{j,k}$ is rational. Extend it by zero outside $2 \leq k \leq 2j$. The first examples are

$$\begin{aligned} C_1(s) &= 2s^2, \\ C_2(s) &= \tfrac{2}{3} s^2 (s+1)(s+2), \\ C_3(s) &= \tfrac{1}{45} s^2 (s+1)(s+2)(4s^2 + 18s + 23), \\ C_4(s) &= \tfrac{1}{315} s^2 (s+1)(s+2)(s+3)(s+4)(2s^2 + 8s + 11). \end{aligned} \tag{3.3}$$

The quadratic coefficient is particularly simple:

$$c_{j,2} = \frac{2H_{2j-1} - H_{j-1}}{j} = \frac{2}{j} \left(H_{2j} - \tfrac{1}{2} H_j \right). \tag{3.4}$$

To express all coefficients in generalized harmonic numbers, put

$$\kappa_\ell^{(N,\alpha)} = (-1)^{\ell-1} (\ell - 1)! \alpha^\ell H_N^{(\ell)}.$$

Expanding the logarithm of each product in (3.2) gives

$$c_{j,k} = \frac{Y_{k-1}(\boldsymbol{\kappa}^{(2j-1,2)}) - Y_{k-1}(\boldsymbol{\kappa}^{(j-1,1)})}{j(k-1)!}, \qquad k \geq 2. \tag{3.5}$$

Only the first $k-1$ entries of each vector are used. Formula (3.5) is an all-order finite compiler, not a recursive instruction to differentiate a large hypergeometric expression.

Theorem 3.1 (All-order harmonic–Hurwitz–Stieltjes identity)**.** *Suppose first that $0 < b < a$. Then, normally for s near zero after its removable pole cancellations,*

$$\mathcal{D}_W(s) = \sum_{d=0}^{D} w_d \sum_{j \geq 1} (-1)^j b^{2j} C_j(s) \zeta(2s + 2j - d, a). \tag{3.6}$$

For every $r \geq 1$,

$$\boxed{\Delta_r(W; a, b) = r! \sum_{d=0}^{D} w_d \sum_{j \geq 1} (-1)^j b^{2j} \mathcal{T}_{j,d,r}(a),} \tag{3.7}$$

where, with $q = 2j - d$,

$$\mathcal{T}_{j,d,r}(a) = \sum_{k=2}^{r} \frac{2^{r-k} c_{j,k}}{(r-k)!} \zeta^{(r-k)}(q, a), \qquad q \neq 1, \tag{3.8}$$

and

$$\mathcal{T}_{j,d,r}(a) = \frac{c_{j,r+1}}{2} + \sum_{k=2}^{r} \frac{(-2)^{r-k} c_{j,k}}{(r-k)!} \gamma_{r-k}(a), \qquad q = 1. \tag{3.9}$$

Empty sums are zero. For arbitrary $b > 0$, choose $N \geq 0$ with $A = a + N > b$. Replace a by A in the double series and add the finite head

$$\sum_{n=0}^{N-1} W(x_n)\{2^{r-1}((-\log(x_n + ib))^r + (-\log(x_n - ib))^r) - (-\log(x_n^2 + b^2))^r\}. \tag{3.10}$$

This is a globally valid normally convergent representation; no small-mass restriction remains after the head is restored.

Proof. The binomial series for the average of $(1 + ib/x)^{-2s}$ and $(1 - ib/x)^{-2s}$ retains exactly the even inverse powers. Their coefficient at x^{-2j} is $(-1)^j b^{2j} (2s)_{2j}/(2j)!$. The quadratic factor has coefficient $(-1)^j b^{2j} (s)_j / j!$. Subtract and sum the resulting powers of x in an initial half-plane to obtain (3.6).

On a compact set of s, the Pochhammer ratios grow at most polynomially in j; their derivatives acquire only polynomial and logarithmic factors. For sufficiently large j, the Hurwitz sum and its fixed order derivatives are bounded by a polynomial in j times a^{-2j}, with constants depending on the compact set and the fixed parameters. Hence $(b/a)^{2j}$ gives normal convergence. The finitely many exceptional Hurwitz factors are continued separately. Their only possible pole at $s = 0$ occurs at $q = 1$, and $C_j(s) = O(s^2)$ cancels it.

For $q \neq 1$, Taylor multiplication directly gives (3.8). At $q = 1$, multiply $C_j(s)$ by

$$\zeta(1 + 2s, a) = \frac{1}{2s} + \sum_{m \geq 0} \frac{(-2)^m \gamma_m(a)}{m!} s^m.$$

The pole raises the required coefficient index from r to $r + 1$, producing (3.9). This proves the derivative formula.

When $b \geq a$, perform exactly the same argument on the tail beginning at $A = a + N$. Its polynomial weight is still $W(x)$ in the same coordinate $x = n + A$. The omitted ordinary finite sum is entire in s; taking its rth derivative gives (3.10). Thus the global formula is an equality of continuations, not an extrapolation of a divergent binomial series. □

3.1 Concrete consequences

Combining (2.15), (3.4), and (3.7) gives the exact harmonic identity

$$\sum_{n \geq 0} \arctan^2\left(\frac{b}{n + a}\right) = \sum_{j \geq 1} \frac{(-1)^{j+1} b^{2j}}{j} \left(H_{2j} - \tfrac{1}{2} H_j\right) \zeta(2j, a), \quad b < a. \tag{3.11}$$

For $a = 1$, its first terms are

$$b^2 \zeta(2) - \tfrac{2}{3} b^4 \zeta(4) + \tfrac{23}{45} b^6 \zeta(6) - \tfrac{44}{105} b^8 \zeta(8) + \cdots.$$

The finite-head version remains valid at every positive b.

For an odd weight, the Stieltjes term is visible already at the second jet. For instance,

$$\Delta_2(x; a, b) = 4b^2 \psi(a) + 4 \sum_{j \geq 2} \frac{(-1)^j b^{2j}}{j} \left(H_{2j} - \tfrac{1}{2} H_j\right) \zeta(2j - 1, a), \quad b < a. \tag{3.12}$$

At the third jet the same resonant term is $24b^2\gamma_1(a)$; at higher orders the compiler produces $\gamma_2(a), \gamma_3(a), \ldots$ with no additional analytic argument. For $W = x^3$ and $r = 2$, the $j = 2$ term is

$$b^4 \left(2 + \tfrac{8}{3}\gamma_0(a)\right).$$

The rational $2b^4$ comes from $c_{2,3}/2$ before multiplication by 2!. Dropping this residue contribution would give an incorrect identity.

The connection to polygamma is explicit: for integers $q \geq 2$,

$$\zeta(q, a) = \frac{(-1)^q}{(q-1)!}\psi^{(q-1)}(a). \tag{3.13}$$

Higher spectral derivatives in (3.8) remain ordinary Hurwitz order derivatives. The formula does not pretend they are all ordinary argument derivatives of polygamma.

4 Noninteger weights and the joint geometry of resonance

For a complex parameter ρ, use $x^\rho = e^{\rho \log x}$ on the positive spectrum. Let $\mathcal{D}(s, \rho)$ be the common-scale linear-minus-quadratic difference with that weight. The meromorphic continuation is most transparently written as

$$\mathcal{D}(s, \rho) = \sum_{j \geq 1} (-1)^j b^{2j} C_j(s) \zeta(2s + 2j - \rho, a), \qquad b < a. \tag{4.1}$$

The same finite head as before globalizes this expression. In particular, normal convergence is locally uniform in both s and ρ away from the explicit polar divisors. The relevant positive odd resonances are $\rho_0 = 2J - 1$, $J \geq 1$.

Theorem 4.1 (Joint meromorphic normal form and finite-part conversion)**.** *Near $(s, \rho) = (0, 2J - 1)$, with $\eta = \rho - (2J - 1)$,*

$$\mathcal{D}(s, \rho) = \frac{(-1)^J b^{2J} C_J(s)}{2s - \eta} + G_J(s, \eta), \tag{4.2}$$

where G_J is jointly holomorphic and $G_J(s, \eta) = O(s^2)$. For a nonresonant fixed ρ, define $\Delta_r(\rho) = \partial_s^r \mathcal{D}(0, \rho)$. Then $\Delta_1(\rho) = 0$. At a resonant weight the fixed-ρ derivative instead satisfies

$$\boxed{\Delta_r(2J - 1) = \mathrm{FP}_{\rho = 2J-1} \Delta_r(\rho) + \frac{r!}{2}(-1)^J b^{2J} c_{J,r+1}.} \tag{4.3}$$

Here the finite part uses $\eta = \rho - (2J - 1)$, and the off-resonance jets are continued meromorphically in that variable. The extra finite term is zero for $r \geq 2J$.

Along a non-tangent ray $\eta = \kappa s$, $\kappa \neq 2$, the first coefficient is

$$\frac{\mathrm{d}}{\mathrm{d}s}\mathcal{D}(s, 2J - 1 + \kappa s)\Big|_{s=0} = \frac{(-1)^J b^{2J} c_{J,2}}{2 - \kappa}. \tag{4.4}$$

The ray $\kappa = 2$ lies on the polar divisor and requires subtraction before restriction.

Proof. In (4.1), exactly the $j = J$ term has a nearby Hurwitz pole. Subtract $1/(2s - \eta)$ from that Hurwitz factor. The remainder is jointly holomorphic. All other factors are locally holomorphic, and normal convergence on a finite-head tail permits the recombination. Every term, including the finite head, starts at s^2, so $G_J = O(s^2)$. This proves (4.2) globally in the positive mass, after the head procedure.

At $\eta \neq 0$, the coefficient of s^r in $C_J(s)/(2s-\eta)$ is

$$-\sum_{k=2}^{r} \frac{2^{r-k} c_{J,k}}{\eta^{r-k+1}}.$$

It has no constant Laurent coefficient in η. At $\eta = 0$, however, the same meromorphic germ restricted before differentiation is $C_J(s)/(2s)$, whose coefficient of s^r is $c_{J,r+1}/2$. The holomorphic remainder gives the same constant term in both orders of operations. Multiplication by $r!(-1)^J b^{2J}$ yields (4.3). Since $\deg C_J = 2J$, the last assertion about vanishing of the local conversion follows. Restricting the normal form to $\eta = \kappa s$ proves (4.4). □

Remark 4.2 (Why a naive interpolation fails)**.** For nonresonant weights the first defect is zero, whereas at positive odd weights it equals the nonzero local coefficient in (2.13). This is not a contradiction to analytic continuation. Differentiation on different slices of a jointly *meromorphic* germ does not give a single jointly holomorphic family across its intersecting divisor. At $\rho = -1$, the individual spectra can themselves have poles at $s = 0$ from their leading term; that leading term cancels in $\mathcal{D}$. At that exceptional weight we discuss derivatives of the common difference, not derivatives of undefined individual values.

The theorem resolves the resonant correction requested for x^ρ. It also identifies exactly which local term is lost if the weight finite part is taken before the spectral derivative. It does not assert holomorphicity on the polar line or exchange two singular limits without a subtraction convention.

5 Logarithmic squares, Gamma integrals, and a dilogarithmic primitive

The simplest nonlocal defect has a useful independent calculus. Let $a > 0$, and let $c, d \in \mathbb{C}$ satisfy $\mathrm{Re}(a+c) > 0$ and $\mathrm{Re}(a+d) > 0$. Put $h = c - d$ and

$$\mathscr{S}(a; c, d) = \sum_{n \geq 0} \big(\log(n+a+c) - \log(n+a+d)\big)^2. \tag{5.1}$$

The summand is $O(n^{-2})$, uniformly on compact sets of these parameters, so this is an ordinary holomorphic sum. In particular $\mathscr{S}(a; ib, -ib) = \Delta_2(1; a, b)$.

Theorem 5.1 (A digamma logarithmic integral)**.** *On the domain above,*

$$\boxed{\mathscr{S}(a; c, d) = 2h \int_0^1 \psi(a+d+ht) \log \frac{t}{1-t}\, \mathrm{d}t.} \tag{5.2}$$

The integral is ordinary and convergent at both endpoints. Moreover,

$$\partial_c \partial_d \mathscr{S}(a; c, d) = -2 \frac{\psi(a+d) - \psi(a+c)}{d-c}, \tag{5.3}$$

with its removable value $-2\psi'(a+c)$ when $c = d$. Every derivative in a is obtained by replacing ψ by the corresponding polygamma in (5.2).

Proof. For each n, integrate the resolvent along the segment from d to c:

$$\log(n+a+c) - \log(n+a+d) = h \int_0^1 \frac{\mathrm{d}t}{n+a+d+ht}.$$

The entire segment lies in the right half-plane after adding a. Compact uniform resolvent estimates justify squaring, summing, and interchanging the two parameter integrals. The standard digamma difference identity gives

$$\sum_{n\ge 0}\frac{1}{(n+A)(n+B)}=\frac{\psi(B)-\psi(A)}{B-A}.$$

Thus $\mathscr{S}$ equals

$$h\int_0^1\int_0^1\frac{f(v)-f(t)}{v-t}\,\mathrm{d}v\,\mathrm{d}t,\qquad f(t)=\psi(a+d+ht).$$

The divided difference has a removable diagonal and is continuous. For a continuously differentiable f, symmetric exclusion of a thin diagonal strip gives

$$\int_0^1\int_0^1\frac{f(v)-f(t)}{v-t}\,\mathrm{d}v\,\mathrm{d}t=2\int_0^1 f(t)\log\frac{t}{1-t}\,\mathrm{d}t.$$

This follows by interchanging the two terms on the symmetric truncated domain and then integrating $1/(t-v)$; the endpoint logarithms are integrable and the omitted divided-difference strip tends to zero. This proves (5.2), including complex parameters.

Differentiating the normally convergent sum in c,d gives $-2\sum((n+a+c)(n+a+d))^{-1}$, which is (5.3). Differentiation under the integral for a fixed number of a derivatives is justified by the same right-half-plane compactness and integrable endpoint logarithms. □

5.1 An anchored primitive with explicit dilogarithms

For the next representation assume real $c>d$ and $a+d>0$, and set $h=c-d>0$. For $y>0$ define

$$f_h(y)=y\log^2(1+h/y)-2h\,\mathrm{Li}_2\!\left(\frac{h}{y+h}\right). \tag{5.4}$$

Direct differentiation, using the ordinary dilogarithm derivative, gives

$$f_h'(y)=\log^2(1+h/y),\qquad f_h(y)=-\frac{h^2}{y}+O(y^{-2})\quad(y\longrightarrow\infty). \tag{5.5}$$

The counterterm in the next theorem is therefore necessary.

Theorem 5.2 (Dilogarithmic and Gamma representations of the primitive)**.** *The series*

$$\mathscr{P}(a;c,d)=h^2\psi(a+d)+\sum_{n\ge 0}\left\{f_h(n+a+d)+\frac{h^2}{n+a+d}\right\} \tag{5.6}$$

is normally convergent for real $a+d>0$ on compact subintervals, and

$$\frac{\mathrm{d}}{\mathrm{d}a}\mathscr{P}(a;c,d)=\mathscr{S}(a;c,d). \tag{5.7}$$

For $a_0+d,a_1+d>0$,

$$\mathscr{P}(a_1;c,d)-\mathscr{P}(a_0;c,d)=2h\int_0^1\log\frac{\Gamma(a_1+d+ht)}{\Gamma(a_0+d+ht)}\log\frac{t}{1-t}\,\mathrm{d}t. \tag{5.8}$$

Thus (5.8) is an exact identity between an ordinary log-Gamma integral and an explicitly convergent dilogarithmic sum.

Proof. The asymptotic formula in (5.5) leaves an $O(n^{-2})$ remainder in (5.6), and differentiating that remainder gives $O(n^{-3})$ on compact sets. Its derivative is

$$h^2\psi'(a+d)+\sum_{n\ge0}\left\{\log^2(1+h/(n+a+d))-\frac{h^2}{(n+a+d)^2}\right\}.$$

Since $\psi'(z)=\sum_{n\ge0}(n+z)^{-2}$, the rational terms cancel and give (5.7). Integrate (5.2) between a_0 and a_1. Fubini is valid on the compact a interval, and the primitive of ψ is the real $\log\Gamma$. This gives (5.8) with the displayed anchoring, not an unspecified additive constant. □

For $|h|<a+d$, the same family has an independent harmonic expansion

$$\mathscr{S}(a;d+h,d)=2\sum_{r\ge2}\frac{(-1)^rH_{r-1}}{r}h^r\zeta(r,a+d). \tag{5.9}$$

This follows from $\log^2(1+z)=2\sum_{r\ge2}(-1)^rH_{r-1}z^r/r$. Its anchored a primitive uses

$$\int_{a_0}^{a_1}\zeta(r,a+d)\,\mathrm{d}a=\begin{cases}\psi(a_1+d)-\psi(a_0+d), & r=2,\\ \dfrac{\zeta(r-1,a_0+d)-\zeta(r-1,a_1+d)}{r-1}, & r>2.\end{cases}$$

This gives another exact bridge between the Gamma integral, harmonic coefficients, and Hurwitz values, and supplies an independent numerical route used in the accompanying tests.

6 Every negative-integer second jet of the bilateral quadratic spectrum

For $0<a<1$, $b>0$, define

$$E(s;a,b)=\sum_{n\in\mathbb{Z}}\bigl((n+a)^2+b^2\bigr)^{-s},\qquad \operatorname{Re}s>\tfrac12. \tag{6.1}$$

Its Poisson representation is

$$E(s;a,b)=\frac{1}{\Gamma(s)}\Bigg\{\sqrt{\pi}\,\Gamma(s-\tfrac12)b^{1-2s}+4\pi^sb^{1/2-s}\sum_{n\ge1}n^{s-1/2}\cos(2\pi na)K_{s-1/2}(2\pi bn)\Bigg\}. \tag{6.2}$$

For completeness, insert $Q^{-s}=\Gamma(s)^{-1}\int_0^\infty t^{s-1}e^{-Qt}\,\mathrm{d}t$ in (6.1). Gaussian Poisson summation converts the lattice heat kernel into $\sqrt{\pi/t}\sum_{k\in\mathbb{Z}}e^{-\pi^2k^2/t}e^{2\pi ika}$. The zero mode integrates to the first term in braces. Pair the positive and negative modes, and use the standard integral $\int_0^\infty t^{\nu-1}e^{-\beta t-\gamma/t}\,\mathrm{d}t=2(\gamma/\beta)^{\nu/2}K_\nu(2\sqrt{\beta\gamma})$ to obtain the second. Initial absolute convergence justifies these steps, and the exponentially convergent Bessel series then continues the result [7].

The expression in braces is holomorphic at every $s=-m$, $m\ge0$. Since $1/\Gamma(s)$ has a simple zero there,

$$E(-m;a,b)=0. \tag{6.3}$$

Write

$$L_m(a,b)=E'(-m;a,b),\qquad M_m(a,b)=E''(-m;a,b).$$

The first jet L_m is the finite family in the predecessor. The second jet differentiates the Bessel order and cannot be obtained merely by squaring or differentiating its already specialized value.

6.1 An explicit half-integer order derivative

For $m \geq 0$ let

$$P_m(y) = \sum_{j=0}^{m} \frac{(m+j)!}{j!(m-j)!2^j} y^j, \qquad A_m(y) = (-1)^m P_m(-y). \tag{6.4}$$

Define polynomials B_m by

$$B_0(y) = 0, \quad B_1(y) = 2y, \quad B_{m+1} = B_{m-1} + (2m+1)yB_m + 2yP_m \quad (m \geq 1). \tag{6.5}$$

All have degree at most m. Let $E_1(x) = \int_x^\infty e^{-t}\,\mathrm{d}t/t$ for $x > 0$.

Lemma 6.1 (Finite polynomial form of the Bessel order derivative)**.** *For $x > 0$,*

$$\partial_\nu K_\nu(x)|_{\nu=m+1/2} = \sqrt{\frac{\pi}{2x}} e^{-x} \left\{ A_m(1/x) e^{2x} E_1(2x) + B_m(1/x) \right\}. \tag{6.6}$$

At $\nu = -m - 1/2$ the derivative has the opposite sign.

Proof. The classical base is $K_{1/2}(x) = k(x) = \sqrt{\pi/(2x)}e^{-x}$. Set $f(x) = e^{2x}E_1(2x)$, so $f' = 2f - 1/x$ and $f'' - 2f' = 1/x^2$. The modified Bessel equation, differentiated in its order at $\nu = 1/2$, is

$$x^2 D'' + xD' - (x^2 + \tfrac{1}{4})D = k.$$

Substituting $D = kf$ reduces it exactly to $x^2(f'' - 2f') = 1$. The real integral representation of K_ν and Laplace localization at its minimum give $(\partial_\nu K_\nu)/k = O(x^{-1})$ for fixed ν as $x \to \infty$; this also follows from the differentiated standard large-x expansion. Our f has the same decay. Their difference solves the homogeneous Bessel equation: the growing solution is excluded by decay, and the remaining multiple of k is excluded by its ratio tending to zero. Thus (6.6) holds at $m = 0$.

Differentiate the order recurrence $K_{\nu+1} = K_{\nu-1} + 2\nu K_\nu/x$. The evenness $K_{-\nu} = K_\nu$ implies oddness of its order derivative. At $\nu = 1/2$, this gives the $m = 1$ polynomial $B_1 = 2y$ and $A_1 = y - 1$. Inductively the differentiated recurrence gives (6.5). The coefficient of $e^{2x}E_1(2x)$ satisfies the same homogeneous recurrence and equals $(-1)^m P_m(-y)$ by the finite recurrence for P_m. This proves the formula at all m. □

The Bessel recurrence and order differentiability are classical [8, 9]; this lemma records a convenient exact normalization for the spectral calculation. The first polynomials are

m	$P_m(y)$	$A_m(y)$	$B_m(y)$
0	1	1	0
1	$1 + y$	$y - 1$	$2y$
2	$1 + 3y + 3y^2$	$1 - 3y + 3y^2$	$2y + 8y^2$
3	$1 + 6y + 15y^2 + 15y^3$	$-1 + 6y - 15y^2 + 15y^3$	$4y + 16y^2 + 46y^3$

6.2 One explicit polylogarithm integral family

For $p \in \mathbb{C}$ set

$$\mathcal{J}_p(a,b) = \int_1^\infty \frac{\mathrm{Li}_p(e^{-2\pi bw + 2\pi ia})}{w+1}\,\mathrm{d}w. \tag{6.7}$$

This is an ordinary absolutely convergent integral, entire in p. For real a and $b > 0$, its series form is

$$\mathcal{J}_p(a,b) = \sum_{n \geq 1} \frac{e^{2\pi ian + 2\pi bn} E_1(4\pi bn)}{n^p}. \tag{6.8}$$

Indeed, expand the polylogarithm under the integral and substitute $t = 2\pi bn(w+1)$ in each exponential integral. For p in a compact set, each p derivative only adds a power of $\log n$; the factor $e^{-2\pi bnw}$ dominates this uniformly when b is bounded below by a positive constant. This proves normal convergence and all claimed interchanges, including parameter differentiation on smaller strips in complex a.

Two exact identities make the added coordinate compatible with the existing calculus:

$$\partial_a \mathcal{J}_p = 2\pi i \mathcal{J}_{p-1}, \qquad \partial_b \mathcal{J}_p = 2\pi \mathcal{J}_{p-1} - \frac{\mathrm{Li}_p(e^{-2\pi b+2\pi ia})}{b}. \tag{6.9}$$

The first follows termwise. For the second, differentiate (6.8) and use $\frac{\mathrm{d}}{\mathrm{d}b}E_1(4\pi bn) = -e^{-4\pi bn}/b$. Every anchored primitive in a is therefore explicit; for example

$$\int_{a_0}^{a_1} \mathcal{J}_p(a,b)\,\mathrm{d}a = \frac{\mathcal{J}_{p+1}(a_1,b) - \mathcal{J}_{p+1}(a_0,b)}{2\pi i}.$$

The integral family is introduced to give a complete expression. No claim that it cannot be eliminated by another identity is made.

6.3 The complete second-jet formula

Let $q = e^{-2\pi b+2\pi ia}$ and, for real p, abbreviate

$$U_p = \mathrm{Re}\,\mathrm{Li}_p(q), \qquad V_p = \mathrm{Re}\,\mathrm{Li}_p^{[1]}(q), \qquad I_p = \mathrm{Re}\,\mathcal{J}_p(a,b).$$

Write $P_m(y) = \sum_j p_{mj}y^j$, $A_m(y) = \sum_j a_{mj}y^j$, $B_m(y) = \sum_j b_{mj}y^j$. Define

$$L_m^{(0)}(b) = -\frac{2^{2m+1}(m!)^2\pi}{(2m+1)!}b^{2m+1}, \qquad c_m(b) = \gamma - H_m + \log(\pi/b). \tag{6.10}$$

Theorem 6.2 (All nonpositive-integer second jets)**.** *For every $m \geq 0$, $0 < a < 1$, and $b > 0$,*

$$\begin{aligned} M_m(a,b) = {} & 4L_m^{(0)}(b)(H_{2m+1} - H_m - \log(2b)) \\ & + \frac{4(-1)^m m! b^m}{\pi^m} \sum_{j=0}^{m} \frac{1}{(2\pi b)^j} \{p_{mj}(c_m(b)U_{m+j+1} - V_{m+j+1}) \\ & \qquad - b_{mj}U_{m+j+1} - a_{mj}I_{m+j+1}\}. \end{aligned} \tag{6.11}$$

Thus a finite set of polylogarithms, their first order derivatives, and $\mathcal{J}_1, \ldots, \mathcal{J}_{2m+1}$ suffices for each second jet.

Proof. Write (6.2) as $E(s) = H(s)/\Gamma(s)$. The reciprocal Gamma expansion at $s = -m + \varepsilon$ is

$$\frac{1}{\Gamma(-m+\varepsilon)} = (-1)^m m! \left\{\varepsilon + (\gamma - H_m)\varepsilon^2 + O(\varepsilon^3)\right\}.$$

Hence

$$M_m = 2(-1)^m m!(H'(-m) + (\gamma - H_m)H(-m)). \tag{6.12}$$

For the zero mode, its logarithmic derivative is $\psi(-m-1/2) - 2\log b$. The recurrence for digamma gives

$$\psi(-m - \tfrac{1}{2}) = -\gamma - 2\log 2 + 2H_{2m+1} - H_m.$$

Substitution in (6.12) yields the first line of (6.11).

For a nonzero Fourier mode, differentiating $\pi^s b^{1/2-s} n^{s-1/2}$ gives $\log(\pi n/b)$. At $s=-m$, $K_{s-1/2}=K_{m+1/2}$ has the finite polynomial P_m. Its order derivative has the minus sign in Lemma 6.1. The resulting mode coefficient, apart from its common prefactor, is

$$e^{-2\pi bn} n^{-m-1} \left\{ P_m((2\pi bn)^{-1})(c_m(b)+\log n) - B_m((2\pi bn)^{-1}) - A_m((2\pi bn)^{-1}) e^{4\pi bn} E_1(4\pi bn) \right\}.$$

Now $\sum q^n \log n/n^p = -\operatorname{Li}_p^{[1]}(q)$, and the last part sums by (6.8). Expanding the three finite polynomials gives (6.11). The exponentially decaying modes and their order derivatives converge normally near $s=-m$, justifying the preceding differentiations and summations. □

The lowest case is already a nontrivial exact identity:

$$\begin{aligned} M_0(a,b) &= 8\pi b\big(\log(2b)-1\big) \\ &\quad + 4\left\{ (\gamma+\log(\pi/b)) \operatorname{Re}\operatorname{Li}_1(q) - \operatorname{Re}\operatorname{Li}_1^{[1]}(q) - \operatorname{Re}\mathcal{J}_1(a,b) \right\}. \end{aligned} \tag{6.13}$$

The formula includes the entire constant Fourier mode. It is not just an oscillatory correction to an unevaluated spectral quantity.

An independent Hurwitz representation is available from the convergent small-mass expansion

$$E(s;a,b) = \sum_{j\ge 0} \frac{(-1)^j (s)_j}{j!} b^{2j} \{ \zeta(2s+2j,a) + \zeta(2s+2j,1-a) \}, \quad b < \min(a,1-a). \tag{6.14}$$

Taylor differentiation at $s=-m$ therefore gives an identity between (6.11) and an ordinary convergent series of Hurwitz jets. In the code this route is implemented independently of the Bessel polynomials and is used to test $m=0,1,2,3$.

7 Harmonic normalization and all anchored mass antiderivatives

Let $u=b^2$ and put $J_{m,r}(a,u) = E^{(r)}(-m;a,\sqrt{u})$, including $J_{m,0}=0$. In the absolute-convergence half-plane and hence by continuation,

$$\partial_u E(s;a,\sqrt{u}) = -sE(s+1;a,\sqrt{u}). \tag{7.1}$$

Differentiating r times at $s=-m$ gives, for $m\ge 1$,

$$\partial_u J_{m,r} = mJ_{m-1,r} - rJ_{m-1,r-1}. \tag{7.2}$$

The lower-jet term cannot be dropped. At the second order it is $-2L_{m-1}$, which explains why the unnormalized second jets are not an Appell-type mass family.

Theorem 7.1 (Gamma-normalized all-order mass transport)**.** *For $m\ge 0$ define the holomorphic spectral germ*

$$\mathfrak{A}_m(\varepsilon;a,u) = \frac{m!\Gamma(1-\varepsilon)}{\Gamma(m+1-\varepsilon)} E(-m+\varepsilon;a,\sqrt{u}) = \frac{E(-m+\varepsilon;a,\sqrt{u})}{\prod_{k=1}^m (1-\varepsilon/k)}. \tag{7.3}$$

Then

$$\partial_u \mathfrak{A}_m = m\mathfrak{A}_{m-1} \quad (m\ge 1). \tag{7.4}$$

Consequently the coefficients $\mathfrak{A}_{m,r} = \partial_\varepsilon^r \mathfrak{A}_m(0;a,u)$ obey $\partial_u \mathfrak{A}_{m,r} = m\mathfrak{A}_{m-1,r}$ for every $r\ge 1$. Their complete harmonic expression is

$$\mathfrak{A}_{m,r} = \sum_{j=0}^{r-1} \binom{r}{j} Y_j(H_m, 1!H_m^{(2)}, \dots, (j-1)!H_m^{(j)}) J_{m,r-j}. \tag{7.5}$$

For every $m \geq 1$ and $u \geq 0$,

$$\mathfrak{A}_{m,r}(a,u) = \sum_{j=0}^{m-1} \binom{m}{j} u^j \mathfrak{A}_{m-j,r}(a,0) + m \int_0^u (u-v)^{m-1} \mathfrak{A}_{0,r}(a,v)\,\mathrm{d}v. \tag{7.6}$$

All integration constants are fixed by

$$J_{m,r}(a,0) = 2^r \{\zeta^{(r)}(-2m,a) + \zeta^{(r)}(-2m,1-a)\}. \tag{7.7}$$

Proof. Use (7.1) at $s = -m + \varepsilon$. Multiplication by the Gamma factor in (7.3) cancels $m - \varepsilon$:

$$\frac{m-\varepsilon}{\prod_{k=1}^{m}(1-\varepsilon/k)} = \frac{m}{\prod_{k=1}^{m-1}(1-\varepsilon/k)}.$$

This proves (7.4) as an identity of holomorphic germs, and coefficient differentiation preserves it.

The multiplier has the convergent germ expansion

$$\prod_{k=1}^{m}(1-\varepsilon/k)^{-1} = \exp\left(\sum_{\ell\geq 1} \frac{H_m^{(\ell)}}{\ell}\varepsilon^\ell\right).$$

The Bell definition (1.2), followed by the product rule, gives (7.5); its last $j = r$ term vanishes because $J_{m,0} = 0$.

The lattice gap $\min(a, 1-a) > 0$ makes the continued spectrum holomorphic in u near zero. One may verify this directly by the normally convergent binomial expansion (6.14), after any needed finite spectral subtractions. At $u = 0$, the spectrum is $\zeta(2s,a)+\zeta(2s,1-a)$, proving (7.7). Repeated integration of the Appell relation from zero, with these initial values, gives (7.6); the integral coefficient is $m!/(m-1)! = m$. Continuation along $u \geq 0$ encounters no zero of any quadratic eigenvalue, so the identity holds for every positive mass. □

The first three normalized orders are

$$\begin{aligned} \mathfrak{A}_{m,1} &= L_m, \\ \mathfrak{A}_{m,2} &= M_m + 2H_m L_m, \\ \mathfrak{A}_{m,3} &= J_{m,3} + 3H_m M_m + 3(H_m^2 + H_m^{(2)})L_m. \end{aligned} \tag{7.8}$$

Here and below L_m, M_m are evaluated at $(a, \sqrt{u})$ when used as functions of u. The first-jet finite expression needed with (6.11) is

$$L_m(a,b) = L_m^{(0)}(b) + 2(-1)^m m! \sum_{j=0}^{m} \frac{(m+j)!\,b^{m-j}}{j!(m-j)!\,4^j \pi^{m+j}} \operatorname{Re}\operatorname{Li}_{m+j+1}(e^{-2\pi b + 2\pi i a}). \tag{7.9}$$

It follows directly from (6.2) and the finite half-integer formula, and agrees with the predecessor.

7.1 Explicit second-order primitives

Since $\mathfrak{A}_{0,2} = M_0$, the $r = 2$ case gives

$$\begin{aligned} &m \int_0^{b^2} (b^2 - v)^{m-1} M_0(a, \sqrt{v})\,\mathrm{d}v \\ &\quad = M_m(a,b) + 2H_m L_m(a,b) - \sum_{j=0}^{m-1} \binom{m}{j} b^{2j} \big(M_{m-j}(a,0) + 2H_{m-j}L_{m-j}(a,0)\big). \end{aligned} \tag{7.10}$$

Every term on the right now has an explicit expression from Theorem 6.2 and (7.9), together with the fixed Hurwitz boundary jets. Thus (7.10) evaluates an infinite family of logarithmic polylogarithm integrals without leaving its integration constants undetermined.

The massless constants can also be written in polylogarithmic order coordinates. Let

$$\mathcal{C}_p(a) = \tfrac{1}{2}(\mathrm{Li}_p(e^{2\pi i a}) + \mathrm{Li}_p(e^{-2\pi i a})),$$

continued in p with the radial convention. For real p this is the real part, but the symmetric expression preserves holomorphy for complex p. The reflected Hurwitz formula [3] gives

$$E(s;a,0) = \frac{4\Gamma(1-2s)}{(2\pi)^{1-2s}} \sin(\pi s)\mathcal{C}_{1-2s}(a) \tag{7.11}$$

near the nonpositive integers. With $k_m = 2(-1)^m(2m)!/(2\pi)^{2m}$, differentiation yields

$$\begin{aligned} L_m(a,0) &= k_m \mathcal{C}_{2m+1}(a), \\ M_m(a,0) &= 4k_m \left\{ (\log(2\pi) + \gamma - H_{2m})\mathcal{C}_{2m+1}(a) - \mathcal{C}^{[1]}_{2m+1}(a) \right\}. \end{aligned} \tag{7.12}$$

At $m = 0$, the first line is $-\log(4\sin^2 \pi a)$. No nonexistent ordinary value of $\mathrm{Li}_1(1)$ is used, since $0 < a < 1$.

8 Two quadratic masses: the exact second derivative and collision calculus

Let $P(y) = \sum_{d=0}^{D} p_d y^d \in \mathbb{C}[y]$ and $u, v > 0$. Define

$$\mathcal{Z}_P(s;a;u,v) = \sum_{n\in\mathbb{Z}} P((n+a)^2)(((n+a)^2+u)((n+a)^2+v))^{-s}, \tag{8.1}$$

$$E_{P,u}(s;a) = \sum_{n\in\mathbb{Z}} P((n+a)^2)((n+a)^2+u)^{-s}. \tag{8.2}$$

Their initial absolute-convergence conditions are $\mathrm{Re}\, s > (2D+1)/4$ and $\mathrm{Re}\, s > D + 1/2$, respectively. We use their meromorphic continuations below. The finite polynomial identity

$$E_{P,u}(s;a) = \sum_{\ell=0}^{D} \frac{P^{(\ell)}(-u)}{\ell!} E(s-\ell; a, \sqrt{u}) \tag{8.3}$$

will reduce the individual weighted jets to the already evaluated family.

8.1 The complete subtraction, not a formal divergent logarithm sum

For $r \geq 2$, put

$$T_r(u,v) = (-1)^r \sum_{k=1}^{r-1} \frac{(u^k - v^k)(u^{r-k} - v^{r-k})}{k(r-k)}. \tag{8.4}$$

These are the coefficients of y^{-r} in $\log^2((y+u)/(y+v))$ at infinity. The first three are

$$\begin{aligned} T_2 &= (u-v)^2, \\ T_3 &= -(u-v)^2(u+v), \\ T_4 &= (u-v)^2 \left\{ \tfrac{2}{3}(u^2+uv+v^2) + \tfrac{1}{4}(u+v)^2 \right\}. \end{aligned} \tag{8.5}$$

The full polynomial part of the weighted logarithmic square is

$$\Pi_P(y;u,v)=\sum_{d=2}^{D}p_d\sum_{r=2}^{d}T_r(u,v)y^{d-r}. \tag{8.6}$$

Define the ordinary sum

$$\mathcal{K}_P(a;u,v)=\sum_{n\in\mathbb{Z}}\left\{P(x^2)\log^2\frac{x^2+u}{x^2+v}-\Pi_P(x^2;u,v)\right\},\qquad x=n+a. \tag{8.7}$$

Its summand is $O(x^{-2})$, locally uniformly in positive u,v. For degree at most one no subtraction is needed. At larger degree the subtraction is generally nonzero and must not be discarded.

Theorem 8.1 (Second derivative of a two-mass spectrum)**.** *Both spectra in* (8.1)–(8.2) *are holomorphic at* $s=0$*. For every polynomial* P*,*

$$\begin{aligned}\mathcal{Z}_P''(0;a;u,v)=2\sum_{\ell=0}^{D}\frac{1}{\ell!}\{P^{(\ell)}(-u)M_\ell(a,\sqrt{u})+P^{(\ell)}(-v)M_\ell(a,\sqrt{v})\}\\ -\mathcal{K}_P(a;u,v).\end{aligned} \tag{8.8}$$

When $u=v$*, the correction vanishes and the formula reduces to* $\mathcal{Z}_P''(0;a;u,u)=4E_{P,u}''(0;a)$*, as required by* $\mathcal{Z}_P(s;a;u,u)=E_{P,u}(2s;a)$*.*

Proof. On either positive lattice tail, all mass expansions and the weight $P(x^2)$ contain only even powers of x. A pole of the associated Hurwitz factor at $s=0$ would require a power x^{-1}, which is impossible. Finite-head subtractions therefore prove holomorphy at zero exactly as in Section 2, without any residue term.

Compare $\mathcal{Z}_P(s)$ with $\frac{1}{2}(E_{P,u}(2s)+E_{P,v}(2s))$. After taking out x^{-4s}, the pointwise difference has zero constant and linear coefficients, and its quadratic coefficient is

$$\tfrac{1}{2}P(x^2)\log^2\frac{x^2+u}{x^2+v}.$$

There is no x^{-1} residue to lower the spectral order. Thus its second derivative is the continued value of the logarithmic-square sum at the fixed lattice coordinate.

The only nonsummable terms of that sum are the polynomial (8.6). Their bilateral regularized sums vanish exactly: for every $k\geq 0$,

$$\zeta(-2k,a)+\zeta(-2k,1-a)=0. \tag{8.9}$$

For example, this follows from $\zeta(-2k,a)=-B_{2k+1}(a)/(2k+1)$ and $B_{2k+1}(1-a)=-B_{2k+1}(a)$, including $k=0$. Consequently the regularized value is exactly the ordinary remainder (8.7), not that remainder plus an unknown constant. Taking the second derivative of the comparison and using (8.3) proves (8.8). □

For example, at $P(y)=y^2$ the full correction is

$$\mathcal{K}_{y^2}(a;u,v)=\sum_{n\in\mathbb{Z}}\left\{x^4\log^2\frac{x^2+u}{x^2+v}-(u-v)^2\right\}.$$

At $P(y)=y^3$ it is

$$\sum_{n\in\mathbb{Z}}\left\{x^6\log^2\frac{x^2+u}{x^2+v}-(u-v)^2x^2+(u-v)^2(u+v)\right\}.$$

These illustrate why positivity of an unsubtracted logarithmic square does not imply positivity of every polynomially regularized correction.

8.2 An ordinary one-dimensional polylogarithmic reduction

The positive first resolvent has the elementary formula

$$E(1;a,\sqrt{t}) = \frac{\pi}{\sqrt{t}} \frac{\sinh(2\pi\sqrt{t})}{\cosh(2\pi\sqrt{t}) - \cos(2\pi a)}. \tag{8.10}$$

One proof takes the difference of the cotangent partial fractions at $a \pm i\sqrt{t}$; another differentiates the known L_0 in the squared mass. Put

$$\mathcal{R}_P(t) = P(-t)E(1;a,\sqrt{t}). \tag{8.11}$$

Theorem 8.2 (The two-mass integral and mixed derivative)**.** *For $u > v > 0$,*

$$\boxed{\mathcal{K}_P(a;u,v) = 2\int_v^u \mathcal{R}_P(t) \log\frac{u-t}{t-v}\,\mathrm{d}t.} \tag{8.12}$$

The endpoint singularities are integrable logarithms. The mixed parameter derivative is

$$\partial_u\partial_v \mathcal{K}_P(a;u,v) = -2\frac{\mathcal{R}_P(u) - \mathcal{R}_P(v)}{v-u}, \tag{8.13}$$

with the continuous coincident value $2\mathcal{R}_P'(u)$. Writing $u = b^2$, $v = c^2$, $b > c > 0$, the integral is equivalently

$$\mathcal{K}_P(a;b^2,c^2) = 4\pi\int_c^b P(-y^2)\left\{1 + 2\,\mathrm{Re}\,\mathrm{Li}_0(e^{-2\pi y+2\pi i a})\right\} \times \log\frac{b^2-y^2}{y^2-c^2}\,\mathrm{d}y. \tag{8.14}$$

For $u < v$ use symmetry; for $u = v$ the correction is zero.

Proof. Polynomial division and (8.9) show that the regularized weighted resolvent is

$$\mathrm{Reg}\sum_{n\in\mathbb{Z}} \frac{P(x^2)}{x^2+t} = P(-t)E(1;a,\sqrt{t}) = \mathcal{R}_P(t).$$

The convention here is precisely polynomial subtraction in x^2. Partial fractions therefore give

$$\mathrm{Reg}\sum_{n\in\mathbb{Z}} \frac{P(x^2)}{(x^2+t)(x^2+w)} = \frac{\mathcal{R}_P(t) - \mathcal{R}_P(w)}{w-t}. \tag{8.15}$$

Its diagonal is removable. Differentiating (8.7) with respect to u, v is valid after subtracting the corresponding polynomial coefficients; the remainder and its parameter derivatives remain $O(x^{-2})$ on compact sets. Since the mixed derivative of the logarithmic square is $-2/((x^2+u)(x^2+v))$, this proves (8.13).

Alternatively write the logarithm as $\int_v^u (x^2+t)^{-1}\,\mathrm{d}t$ and square it. Polynomial subtraction commutes with these integrals coefficient by coefficient; the remaining uniformly convergent sum may be integrated by ordinary Fubini. Thus $\mathcal{K}_P$ is the double integral of (8.15) over $[v,u]^2$. The same symmetric divided-difference calculation used in Theorem 5.1, with its sign reversed, gives (8.12).

Finally substitute $t = y^2$ and use the absolutely convergent Poisson kernel identity

$$\frac{\sinh(2\pi y)}{\cosh(2\pi y) - \cos(2\pi a)} = 1 + 2\,\mathrm{Re}\,\frac{e^{-2\pi y+2\pi i a}}{1 - e^{-2\pi y+2\pi i a}} = 1 + 2\,\mathrm{Re}\,\mathrm{Li}_0(e^{-2\pi y+2\pi i a}).$$

This proves (8.14) and its convergence. □

The formulas give the full second product derivative in terms of finite negative-integer second jets and a displayed ordinary integral. There is no implicit subtraction of two divergent logarithmic sums.

8.3 Every collision coefficient is a finite polygamma expression

Theorem 8.3 (Convergent collision expansion)**.** *Let $g = \operatorname{dist}(a, \mathbb{Z}) = \min(a, 1-a)$. For $u > 0$ and $|h| < u + g^2$, the correction has the convergent Taylor expansion*

$$\mathcal{K}_P(a; u+h, u) = 2\sum_{r\geq 2} \frac{(-1)^r H_{r-1}}{r} h^r \sum_{\ell=0}^{\min(D,r-1)} \frac{P^{(\ell)}(-u)}{\ell!} E(r-\ell; a, \sqrt{u}). \tag{8.16}$$

It defines the analytic branch continued from $h = 0$ in that disk. In particular,

$$\lim_{h\to 0} \frac{\mathcal{K}_P(a; u+h, u)}{h^2} = P(-u)E(2; a, \sqrt{u}) + P'(-u)E(1; a, \sqrt{u}) = -\mathcal{R}_P'(u). \tag{8.17}$$

All resolvents on the right of (8.16) have positive integer order and admit finite polygamma expressions.

Proof. The elementary identity

$$\log^2(1+z) = 2\sum_{r\geq 2} \frac{(-1)^r H_{r-1}}{r} z^r, \qquad |z| < 1,$$

follows either by multiplying the logarithm series or by differentiating both sides. Put $z = h/(x^2+u)$. Its modulus is uniformly less than one on every compact subdisk $|h| < u + g^2$. The polynomial subtraction in (8.7) depends holomorphically on h, and its remainder is normally summable on such compact sets. Taylor coefficients may therefore be extracted after summation and subtraction.

At order r, polynomial expansion around $-u$ gives

$$\frac{P(x^2)}{(x^2+u)^r} = \sum_{\ell=0}^{D} \frac{P^{(\ell)}(-u)}{\ell!} (x^2+u)^{\ell-r}.$$

For $\ell \geq r$ the terms are polynomials in x^2 and have zero bilateral regularized sum. For $\ell < r$ they are ordinarily summable positive-order resolvents. This gives exactly (8.16). The $r = 2$ coefficient yields the first equality in (8.17); differentiating (8.11) and using $\partial_u E(1) = -E(2)$ gives the second. □

For clarity, a finite polygamma formula can be written without a recursive mass derivative. For $N \geq 1$, define

$$A_{N,r}(b) = (-1)^{N-r} \binom{2N-r-1}{N-r} (-2ib)^{-(2N-r)}, \quad 1 \leq r \leq N,$$
$$\Xi_1(z) = -\psi(z), \qquad \Xi_r(z) = \frac{(-1)^r}{(r-1)!} \psi^{(r-1)}(z) \quad (r \geq 2). \tag{8.18}$$

Then the elementary partial fractions at the conjugate roots give

$$E(N; a, b) = 2\,\mathrm{Re} \sum_{r=1}^{N} A_{N,r}(b)\{\Xi_r(a+ib) + \Xi_r(1-a+ib)\}. \tag{8.19}$$

For $r = 1$ the two conjugate harmonic tails cancel before the digamma values are taken. The formula can also be checked from

$$E(N; a, \sqrt{u}) = \frac{(-1)^{N-1}}{(N-1)!} \partial_u^{N-1} E(1; a, \sqrt{u}). \tag{8.20}$$

Thus no noninteger-order or unevaluated infinite spectral sum is required in any collision coefficient.

For $P = 1$, the expansion begins

$$\mathcal{K}_1(a; u+h, u) = h^2 E(2) - h^3 E(3) + \tfrac{11}{12} h^4 E(4) - \tfrac{5}{6} h^5 E(5) + \cdots, \tag{8.21}$$

all at $(a, \sqrt{u})$. For higher-degree P, the extra translated polynomial derivatives in (8.16) are essential; keeping only $P(-u)E(r)$ would be incorrect.

8.4 An exact averaged logarithmic-square identity

An explicit elementary consequence provides both an independent check and a useful integral evaluation.

Corollary 8.4 (Average over the lattice shift)**.** *For $b, c > 0$,*

$$\int_0^1 \mathcal{K}_1(a; b^2, c^2)\,\mathrm{d}a = \int_{-\infty}^{\infty} \log^2 \frac{x^2+b^2}{x^2+c^2}\,\mathrm{d}x = 8\pi\{b\log(2b) + c\log(2c) - (b+c)\log(b+c)\}. \tag{8.22}$$

Consequently

$$\int_0^1 \mathcal{Z}_1''(0; a; b^2, c^2)\,\mathrm{d}a = 8\pi\{b\log(2b) + c\log(2c) + (b+c)\log(b+c) - 2(b+c)\}. \tag{8.23}$$

Proof. Absolute convergence unfolds the shift average of the sum into the real-line integral. Alternatively the mean of $E(1; a, \sqrt{t})$ is $\pi/\sqrt{t}$. Averaging (8.12) and substituting $t = y^2$ gives, for $b > c$,

$$4\pi \int_c^b \log \frac{b^2-y^2}{y^2-c^2}\,\mathrm{d}y.$$

Split the logarithm into its four linear factors. The integrals of $\log(b-y)$ and $-\log(y-c)$ cancel. Integrating the remaining $\log(b+y) - \log(y+c)$ gives twice the braces in (8.22). Symmetry covers $c > b$ and continuity covers $b = c$. The mean of (6.13) is its zero Fourier mode $8\pi b(\log(2b) - 1)$, so (8.8) proves (8.23). □

9 Source audit, mathematical status, and reproducible evidence

9.1 What was actually inspected

The canonical manuscript README, its directory structure, the incoming directory listing, and repository tree metadata were retrieved through the connected GitHub tools. The tree commit returned during this reading was

`73f18ef013d5b74eafc883efb2c15210642df12a`

The incoming catalogue contained six archives. Their names and Git blob identifiers are recorded in `PROVENANCE.json`. Their binary members were not inspected: direct binary retrieval was unavailable in this session. It would therefore be incorrect to claim a full audit of those archives or a proof that none contains overlapping results.

The complete matching-title TeX of `ProveIt_Quadratic_Spectrum_Stieltjes_2026-10-10.tex` was recovered from the user's Library and materialized for programmatic inspection. The targeted proof sections, finite first-jet formulas, and final research questions were read. Its SHA-256 is

`4185cc4aa2299c8f417ca376b3ddc4af3fab58752d4cd93f70c2afad661f95aa`

The predecessor records its own earlier repository pin; that pin is not replaced by the newer metadata anchor above. Matching the title does not prove byte identity with the incoming ZIP. The source hash is recorded so a future repository integration can resolve that provenance question exactly.

The source's higher-jet warning was correct. No new error in the inspected first-jet proof is asserted here. The work extends that proof rather than silently replacing it. The canonical Gaussian S_6 and revised S_8 conjectures are not settled by these spectral identities, and their status must not be changed on the basis of this report.

9.2 Corrections to tempting but invalid extensions

The following are safeguards demonstrated by the present formulas, not accusations that the predecessor made these mistakes.

First-jet parity is not higher-jet multiplicativity. Even W eliminates the residue coefficients R_k, but not the regular terms. Formula (2.15) disproves the extension already at $W = 1$, $r = 2$; Theorem 2.3 settles universal higher-order vanishing completely.

The colliding coefficient is one order higher. At $2j - d = 1$, the Hurwitz pole turns $c_{j,r+1}s^{r+1}$ into a contribution of order s^r. The term $c_{j,r+1}/2$ in (3.9) cannot be dropped. Weight finite parts and spectral derivatives differ by exactly (4.3).

Finite heads matter beyond the first derivative. The ordinary head in (3.10) vanishes for $r = 1$, but usually not for $r \geq 2$. A calculation that merely replaces a by $a + N$ in the tail loses an explicit finite term. The tests compare multiple head choices at masses both inside and outside the original small-mass disk.

The polynomial subtraction is part of the two-mass answer. At degree at least two, the raw logarithmic-square sum diverges. Equation (8.6) gives every required term, and (8.9) explains why no further finite constant is present. The second derivative of a product is not the sum of its two second derivatives; the scale factor and the correction are both shown in (8.8).

A sufficient function family is not a minimal one. The integral $\mathcal{J}_p$ makes the second-jet formula complete. Neither a failed reduction search nor the appearance of E_1 proves that $\mathcal{J}_p$ cannot be reduced further. That minimality question remains separate.

9.3 Executed finite exact checks

The standalone script `verification/check_exact.py` completed **834 exact assertions** using rational arithmetic and SymPy. It also rejected **five deliberately corrupted identities**. The exact suite independently compares the Pochhammer and harmonic coefficient constructions through $j = 20$, checks the Bessel and order-differentiated differential equations through $m = 10$, checks Gamma-normalization identities through $m = 16$, verifies logarithmic-square coefficients through degree 18, verifies polynomial resolvent remainders through weight degree 10, and checks the relevant Bernoulli reflection and first-anomaly coefficients.

The report `data/exact_results.json` enumerates every assertion rather than reporting an inferred count. The exact rational table `data/coefficient_table.json` gives low-order C_j, all checked P_m, A_m, B_m, and sample two-mass subtraction coefficients. These are finite regression certificates. The mathematical range beyond the checked indices follows from the proofs, not from extrapolation.

9.4 Executed numerical diagnostics

Three separately replayable suites completed **63 comparisons** at 50 decimal working digits with scaled tolerance 10^{-34}. The retained maximum scaled discrepancy is below $8 \cdot 10^{-49}$. The factorization suite has 19 comparisons, the Fourier/Bessel suite 24, and the collision suite 20. Their individual inputs, outputs, and discrepancies are in the three corresponding JSON files under `data/`.

The independent comparisons include separate complex-shift Hurwitz jets versus the cancellation compiler, global digamma integrals versus the head/tail formula, Bessel order derivatives versus finite polynomials, ordinary polylogarithm integrals versus exponential-integral series, and the negative-integer

second jets versus the independent quadratic binomial expansion. The collision tests compare subtracted spectral sums with ordinary integrals and Taylor expansions. Three additional comparisons compute the two-mass second derivative directly from its double Pochhammer expansion, independently of the logarithmic-square decomposition.

These are not rigorous interval enclosures. The truncation routines choose generous precision-dependent cutoffs, and quadrature is floating-point. The implementation avoids a numerical library overflow at astronomically small complex polylogarithm arguments by using the leading term of their convergent series once the discarded terms are far below the diagnostic precision. That engineering safeguard does not replace an error-certified numerical analysis. No proof-assistant formalization or external peer review is claimed.

9.5 Replay and additive integration

From the package root, with Python, mpmath, SymPy, and a LaTeX engine:

```
python verification/verify_manifest.py
python verification/check_exact.py
python verification/check_numeric.py --part factorization \
  --output data/numerical_factorization.json
python verification/check_numeric.py --part fourier \
  --output data/numerical_fourier.json
python verification/check_numeric.py --part collision \
  --output data/numerical_collision.json
python verification/build.py
```

Run manifest verification before replaying or rebuilding. The delivered JSON files are receipts of executed runs. Replaying with other output paths preserves those receipts. The build script assembles a self-contained TeX source from the modular sources and runs three serial LaTeX passes. Rebuilding can change PDF metadata; manifest verification is for the delivered bytes, not a certification of every future rebuild.

A suitable additive destination is `Analysis/Polylogarithms/docs/reports/higher-quadratic-jets/`. The integration notes map each theorem to the predecessor's specific research question. All manuscript labels use the prefix `hqj:`. The short integration fragment is an overview with pointers, not a substitute for the full proofs. No remote commit, overwrite, or automatic conjecture-status change is part of this delivery.

10 Further research questions and concrete next targets

The questions below go beyond the settings proved here. They are formulated to keep exact evaluation and analytic validity separate from unsupported claims of transcendental nonreduction.

1. **Third and higher negative-integer Fourier jets.** For $r \geq 3$, derive a terminating polynomial recurrence for the order derivatives $\partial_\nu^{r-1} K_\nu$ at half-integers. Determine a finite list of explicit convergent polylogarithmic integral kernels at each fixed r and give every coefficient of $E^{(r)}(-m; a, b)$ uniformly in m. The present $\mathcal{J}_p$ is the first additional coordinate; the normalized mass ladder supplies consistency relations for any proposed higher-order answer.

2. **Eliminating or identifying the added integral coordinate.** At rational a or special b, seek exact identities for $\mathcal{J}_p(a, b)$ in terms of familiar polylogarithmic, Hurwitz, or Gamma order derivatives. Alternatively define a formal function algebra precisely enough to prove a limited

obstruction there. Do not infer numerical nonreducibility merely from the failure of a finite symbolic basis search.

3. **Higher-order two-mass and many-mass interactions.** Replace the second-jet logarithmic square by the cubic and quartic mixed logarithmic terms of a product of several quadratic factors. Derive explicit polynomial subtractions and reduce their parameter integrals to divided-resolvent kernels. A first useful target is the complete third jet for two masses, with a convergent collision expansion and correctly normalized repeated-mass limit.

4. **Several noninteger weight resonances at once.** The local equation $2s - \eta = 0$ in Theorem 4.1 has one simple pole. For products with independently varied orders and several weight parameters, classify the intersecting polar divisors and give a multivariable finite-part conversion law. State explicitly which transverse coordinates define the finite part and how that choice affects the local correction.

5. **Rational-shift arithmetic projections.** For $a = p/q$, apply finite character projectors simultaneously to the harmonic–Stieltjes compiler, the second-jet Fourier family, and the two-mass integral. Produce a complete small-conductor dictionary with even and odd character components kept distinct. The goal is an exact family of Gamma or Dirichlet-L derivative identities, not a numerical guess about the number of independent periods.

6. **Mixed shift–mass primitives.** Combine the a-primitive law (6.9) with the normalized mass ladder to evaluate rectangular parameter integrals of the second jets. Determine which mixed primitives close in the same finite set of Li_p, $\mathrm{Li}_p^{[1]}$, and $\mathcal{J}_p$, and which require an additional explicitly anchored coordinate.

7. **One-sided polynomially weighted logarithmic products.** The bilateral cancellation of even polynomial sums simplifies Section 8. On a one-sided lattice those polynomial values need not vanish. Derive the full two-mass analogue, with its Bernoulli values, digamma terms, and local Stieltjes corrections retained. The finite subtraction framework in Section 2 already gives a precise convention for this extension.

8. **Analytic formalization and certified evaluation.** Formalize the common-scale pole cancellation, the finite-head normal-convergence lemma, and the logarithmic divided-difference integral identity in a proof assistant. Separately implement interval arithmetic for the $\mathcal{J}_p$ integrals and the globally subtracted collision sums. The existing exact coefficient tables and mutation controls are finite fixtures, not substitutes for those analytic developments.

Conclusion

The key distinction is between a local residue and the full higher spectral derivative. First-order even-weight factorization survives because the residue vanishes; higher derivatives retain a normally convergent logarithmic contribution. The all-order compiler makes that contribution and every Stieltjes contact term explicit. A separate half-integer Bessel calculation completes every negative-integer second jet, and a Gamma normalization restores exact mass integration at all spectral orders. Finally the two-mass product has a complete ordinary integral correction and a convergent, finite-polygamma collision calculus. These are exact identities with stated domains and proofs, not conjectural numerical reductions.

A Independent Hurwitz expansions used in the verification

This appendix gives the separate analytic formulas used for the most important numerical comparisons. They are useful identities in their own right and make the independence of the implementations inspectable. Let

$$R_a(z) = \zeta(z,a) + \zeta(z,1-a), \qquad g = \min(a, 1-a).$$

All primes in this appendix mean derivatives of R_a in z.

A.1 The second jet at a negative integer

For fixed $m \geq 0$ and $0 < b < g$, define for $0 \leq j \leq m$

$$h_{mj} = H_m - H_{m-j}, \qquad h^{(2)}_{mj} = H^{(2)}_m - H^{(2)}_{m-j}, \qquad q_j = 2j - 2m.$$

Differentiating (6.14) gives

$$\begin{aligned} M_m(a,b) = &\sum_{j=0}^{m} \binom{m}{j} b^{2j} \left\{ (h_{mj}^2 - h^{(2)}_{mj}) R_a(q_j) - 4h_{mj} R_a'(q_j) + 4R_a''(q_j) \right\} \\ &+ \sum_{j=m+1}^{\infty} \frac{(-1)^{j+m} b^{2j}}{j\binom{j-1}{m}} \left\{ 2(H_{j-m-1} - H_m) R_a(q_j) + 4R_a'(q_j) \right\}. \end{aligned} \tag{A.1}$$

For $j \leq m$, logarithmic differentiation of $(s)_j$ at $s = -m$ gives the finite harmonic differences. For $j > m$, separate its simple factor $s + m$ and use

$$\frac{(-1)^j(-m+\varepsilon)_j}{j!} = \frac{(-1)^{j+m}}{j\binom{j-1}{m}} \left\{ \varepsilon + (H_{j-m-1} - H_m)\varepsilon^2 + O(\varepsilon^3) \right\}.$$

This proves (A.1). There are no colliding Hurwitz poles at q_j, since every q_j is even. Comparing this formula with (6.11) gives a proved Hurwitz–polylogarithm integral identity for every m, independent of numerical testing.

A.2 The two-mass second derivative without using the defect

For $0 < u, v < g^2$, expand the two quadratic factors independently:

$$\mathcal{Z}_P(s;a;u,v) = \sum_{d=0}^{D} p_d \sum_{j,k\geq 0} \frac{(-1)^{j+k}(s)_j(s)_k}{j!k!} u^j v^k R_a(4s + 2j + 2k - 2d). \tag{A.2}$$

Normal convergence follows from the geometric factors relative to the lattice gap. At $s = 0$, all potential Hurwitz arguments are even, so no pole correction is required. Collecting by $\ell = j + k$ yields

$$\begin{aligned} \mathcal{Z}_P''(0;a;u,v) = &\, 16 \sum_{d=0}^{D} p_d R_a''(-2d) \\ &+ \sum_{\ell\geq 1} (-1)^\ell \sum_{d=0}^{D} p_d \Bigg[\frac{u^\ell + v^\ell}{\ell} \left\{ 2H_{\ell-1} R_a(2\ell - 2d) + 8R_a'(2\ell - 2d) \right\} \\ &\qquad + 2R_a(2\ell - 2d) \sum_{j=1}^{\ell-1} \frac{u^j v^{\ell-j}}{j(\ell - j)} \Bigg]. \end{aligned} \tag{A.3}$$

The $j = k = 0$ term contributes the first line. Terms with exactly one positive index use $(s)_\ell/\ell! = s/\ell + H_{\ell-1}s^2/\ell + O(s^3)$; terms with both indices positive use their two linear coefficients. This proves (A.3) directly from the defining product spectrum. The separate code route does not call $\mathcal{K}_P$, the Bessel recurrence, or the product-defect theorem. Its agreement with (8.8) is therefore a useful cross-check of signs and scale normalization.

B Selected symbolic and numerical receipts

The delivered receipts include all individual checks. Two sample values from the executed factorization and second-jet diagnostics are

$$\sum_{n\geq 0} \arctan^2\left(\frac{0.4}{n+0.7}\right) \approx 0.39398901850580706937356548661078543355 92,$$

$$\Delta_2(1; 0.7, 0.4) \approx -1.57595607402322827749426194644314173423 7.$$

These displayed decimals illustrate the normalization. They are not claimed as interval-enclosed values. The actual test files retain 50-digit working-precision values and scaled errors.

The rational constructions also detect simple implementation errors exactly. Replacing $C_1(s) = 2s^2$ by $3s^2$ fails the Pochhammer coefficient comparison; omitting $B_1(y) = 2y$ fails the differentiated Bessel equation; omitting the lower-jet term in (7.2) fails the Gamma-normalization relation. The final negative controls are intentionally false statements, not failed mathematical claims retained as theorems.

References

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